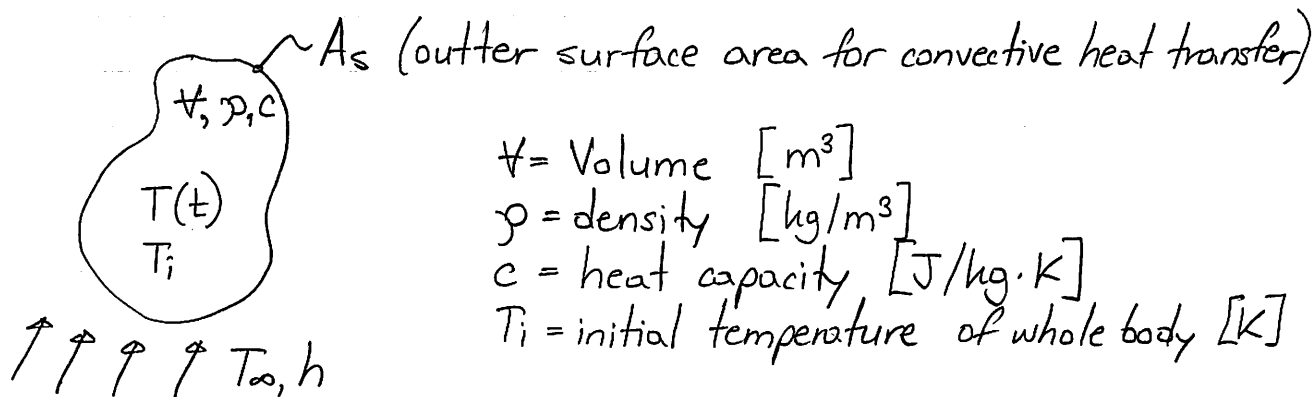


Lumped Capacitance Analysis (Transient Problems)

This analysis method is very valuable for a whole host of problems



V = Volume $[m^3]$

ρ = density $[kg/m^3]$

c = heat capacity $[J/kg \cdot K]$

T_i = initial temperature of whole body $[K]$

Doing an energy balance on our body of interest:

$$\underset{0}{\overset{\nabla}{E}}_{in} - E_{out} + \underset{0}{\overset{\nabla}{E}}_{gen} = E_{stored}$$

$$-hA_s(T - T_\infty) = \frac{\partial}{\partial t}(\rho V c T) \Rightarrow \text{Assume constant properties}$$

$$\rho V c \frac{\partial T}{\partial t} + hA_s(T - T_\infty) = 0$$

$$\frac{\partial T}{\partial t} + \frac{hA_s}{\rho V c}(T - T_\infty) = 0 \Rightarrow \text{Let } \theta = T - T_\infty$$

$$\frac{\partial \theta}{\partial t} + \frac{hA_s}{\rho V c}\theta = 0 \Rightarrow \text{Let } \lambda = \frac{hA_s}{\rho V c}$$

$$\frac{\partial \theta}{\theta} = -\lambda dt$$

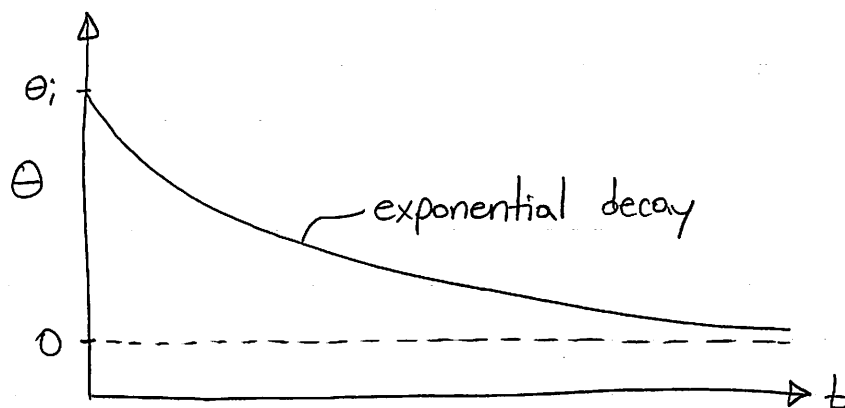
$$\ln \theta = -\lambda t + C_2$$

$$e^{\ln \theta} = e^{-\lambda t + C_2} \Rightarrow \theta(t) = C_3 e^{-\lambda t}$$

Our B.C. or I.C. is: $\Theta(t=0) = \Theta_i = T_i - T_\infty$

$$\Theta(t=0) = C_3 \underbrace{e^{-\cdot(0)}}_1 = \Theta_i \Rightarrow \boxed{C_3 = \Theta_i}$$

$$\boxed{\Theta(t) = \Theta_i e^{-\frac{hA_s t}{\rho c}}} \Rightarrow \text{Lumped capacitance model. Exponential decay in body temp.}$$



Note, the lumped capacitance model is only valid for bodies that change temperature uniformly as they cool or heat up.

$$\boxed{Bi < 0.1}$$

Remember, for a slab: $Bi_t = \frac{h \overset{\substack{\uparrow \\ t = \text{slab thickness not time!}}}{t}}{k} \leq 0.1$

What about a cylinder or sphere? What thickness do we use in our Bi ?

$$\overset{\substack{\uparrow \\ \text{Sometimes labeled } L_c \text{ for length scale.}}}{t} = \frac{\text{Volume}}{\text{Surface Area}}$$

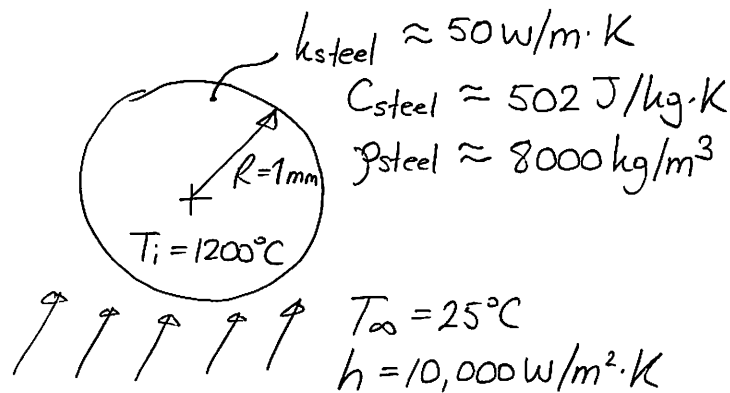
Sometimes labeled L_c for length scale.

$$\Rightarrow \text{Slab} \Rightarrow \frac{t \cdot 2L \cdot W}{2(L \cdot W)} = t; \quad \boxed{Bi_{\text{slab}} = \frac{ht}{k}}$$

$$\text{Cylinder} \Rightarrow \frac{\pi R^2 L}{2\pi R L} = \frac{R}{2}; \quad \boxed{Bi_{\text{cyl}} = \frac{hR}{2k}}$$

$$\text{Sphere} \Rightarrow \frac{\frac{4}{3}\pi R^3}{4\pi R^2} = \frac{R}{3}; \quad \boxed{Bi_{\text{sph}} = \frac{hR}{3k}}$$

Example | A steel quenching process for ball bearings requires the ball bearing to reach 100°C in order to work. The initial ball temperature is 1200°C , and it is dipped in cold water at 25°C . The ball radius is $R = 1\text{mm}$. The heat transfer coefficient is $h = 10,000\text{ W/m}^2\cdot\text{K}$. Find how long the quenching time should be.



First step: Check the $B_{\text{sphere}} < 0.1$

$$B_{\text{sph}} = \frac{hR_{\text{sph}}}{k} = \frac{(10,000\text{ W/m}^2\cdot\text{K})(0.001\text{m})}{3(50\text{ W/m}\cdot\text{K})} = 0.066 < 0.1$$

$\therefore B_{\text{sph}} \leq 0.1 \Rightarrow \text{Lumped Capacitance OK to use.}$

We just solved that:

$$\theta(t) = T - T_{\infty} = (T_i - T_{\infty}) e^{-\frac{hA_s t}{\rho C V}}$$

$$\frac{100^{\circ}\text{C} - 25^{\circ}\text{C}}{1200^{\circ}\text{C} - 25^{\circ}\text{C}} = e^{-\frac{(10,000)(3)t}{(0.001)(8000)(502)}}$$

$$0.064 = e^{-7.47t}$$

$$\ln(0.064) = -7.47t$$

$$t = 0.375$$

If we look back at our solution for lumped capacitance

$$\underbrace{\frac{\theta}{\theta_i}}_{\text{Dimensionless}} = \exp \left[\underbrace{-\frac{hA_s t}{\rho V c}}_{\text{Dimensionless}} \right]$$

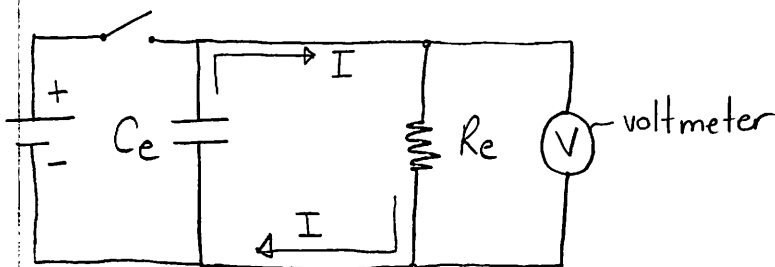
Dimensionless $\Rightarrow$ Implies that:

$$\tau = \underbrace{\left(\frac{1}{hA_s} \right)}_{\substack{\text{Convective} \\ \text{Resistance} \\ \equiv R_t}} \underbrace{(\rho V c)}_{\substack{\text{Thermal} \\ \text{Capacitance} \\ \equiv C_t}} = R_t \cdot C_t \quad (\text{Thermal time constant})$$

$\Downarrow$
The larger τ is, the slower the response of the body.

Any increase in R_t or C_t will cause a solid to respond more slowly to changes in its thermal environment.

Analogous to the voltage decay when discharging a capacitor through a resistor in an RC circuit.



$$V(t) = V_0 e^{-\frac{t}{R_e C_e}} \quad \leftarrow \text{solution}$$

$$V = I R_e \quad (\text{Voltage})$$

$$\underbrace{C_e \frac{dV}{dt}}_{\substack{\text{Current at} \\ \text{capacitor}}} + \underbrace{\frac{V}{R_e}}_{\substack{\text{Current through} \\ \text{resistor}}} = 0$$

②

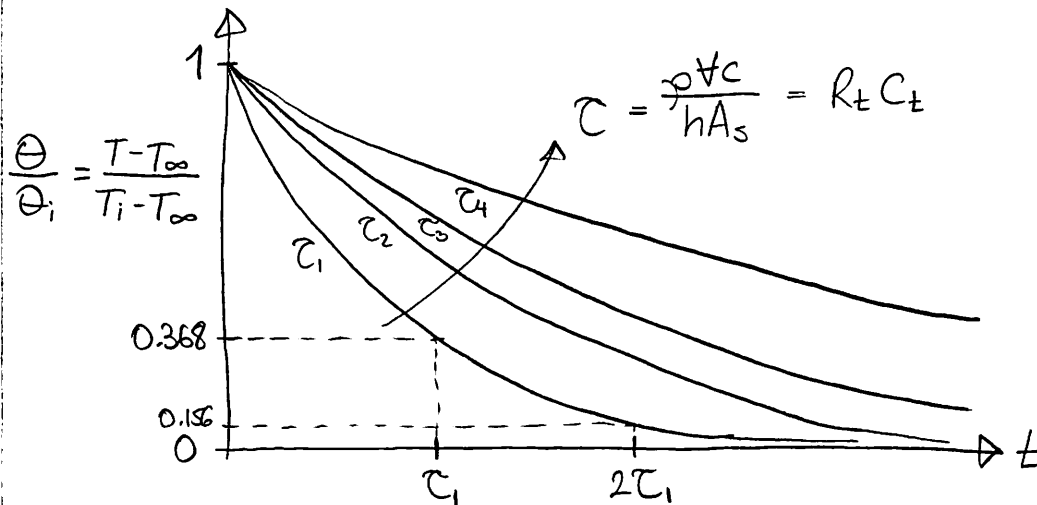
- The larger the capacitance C_e , the more energy stored and the slower the voltage decay

③

- The larger the thermal capacitance $(\rho V c)$, the more energy stored and the slower the temperature decay

- The larger the resistor (electrical or thermal), the smaller the current (electrical or thermal) and the slower the temp decay

We can now draw our solutions in dimensionless form:



We can take the analysis one step further and say:

$$\frac{hA_s t}{\rho V c} = \frac{ht}{\rho c L_c} \Rightarrow L_c \equiv \frac{V}{A_s} \text{ (length scale)}$$

$$= \frac{hL_c}{k} \cdot \frac{k}{\rho c} \cdot \frac{t}{L_c^2} = \underbrace{\frac{hL_c}{k}}_{Bi} \cdot \underbrace{\frac{\alpha t}{L_c^2}}_{Fo} \Rightarrow \boxed{\frac{\Theta}{\Theta_i} = e^{-Bi Fo}}$$

↳ Lumped capacitance analysis in dimensionless form.

$$\boxed{Fo \equiv \text{Fourier \#} = \frac{\alpha t}{L_c^2}} \Rightarrow \text{Dimensionless time that characterises transient heat transfer prob.}$$

$$\boxed{\alpha = \text{Thermal diffusivity} = \frac{k}{\rho c}} \Rightarrow \text{Describes how quickly a material responds to a change in temperature. (Thermal inertia)}$$

Note:

$$\boxed{Fo \equiv \frac{\text{Diffusive transport rate}}{\text{Storage rate}}}$$

$$\begin{aligned} Fo &= \frac{\alpha t}{L_c^2} = \frac{k t}{\rho c L_c^2} \cdot \left(\frac{\Delta T}{\Delta T} \right) \\ &= \left(\frac{k \Delta T}{L} \right) / \left(\frac{\rho c \Delta T L}{t} \right) \\ &= q''_{\text{cond}} / q''_{\text{stored}} \end{aligned} \quad (59)$$